XRPL Validator — Daily Engineering Report · 2026-04-21

Period: 2026-04-21 00:00 to 24:00 US Eastern (2026-04-21 04:00 to 2026-04-22 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 6 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-04-21): 0.99612 (83 missed of 21,372). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(ffi_engine): OverlayedDbProvider::succ() must consult overlay, not just snapshot** (`69bc726`, xrpl-validator) The previous implementation iterated only the RocksDB snapshot, so when an earlier transaction in the same ledger modified or deleted an SLE in a directory the current transaction walks (Offer-book traversal in particular), libxrpl saw a stale RocksDB neighbor and crossed wrong offers. Cascade: wrong mutations -> wrong sequence -> tefPAST_SEQ on downstream transactions from the same accounts.
- **fix(ffi_engine): tefBAD_LEDGER recovery — fetch missing SLE and retry apply** (`6dd8f22`, xrpl-validator) When AccountDelete walks the deleted account's owner directory, libxrpl FFI returns `tefBAD_LEDGER` with a fatal message like: "DeleteAccount: directory node in ledger N has index to object that is missing: <64-hex-key>" meaning state.rocks is missing an SLE referenced by the owner dir (bulk_sync / ws_sync completeness gap). Historically these divergences also cascaded: 100% of observed Payment terPRE_SEQ …
- **feat(dashboard): /logs page — progress journal + live GitHub changelog** (`8176931`, xrpl-validator) Adds a new /logs route with two tabs: - PROGRESS — hand-written narrative sourced from PROGRESS.md at the repo root. Read at request time via fs.readFileSync so edits to the markdown show up without a rebuild. Clearly labeled as manually-curated.
- **fix(ffi_engine): loop tefBAD_LEDGER recovery for multi-missing-SLE owner dirs** (`4ab09ec`, xrpl-validator) The single-try recovery shipped in 6dd8f22 handles AccountDeletes with exactly one missing SLE in the owner directory. When the dir references multiple missing SLEs (NFTokenOffers + Offers + RippleStates on an old high-activity account), the first retry injects SLE #1 but libxrpl then trips on SLE #2 and returns tefBAD_LEDGER again.

Tests and campaigns

- **test(node): add release-safety guard against shipped seeds, tokens, and non-local IPs** (`cbda593`, xrpl-validator)

Docs and operations

- **docs(progress): redact file:line for unfixed SEV findings** (`a59ae04`, xrpl-validator)

Independence program

No independence record for UTC day 2026-04-21; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
04-21 05:40	—	xrpl-validator	`cbda593`	test(node): add release-safety guard against shipped seeds, tokens, and non-local IPs
04-21 06:10	—	xrpl-validator	`69bc726`	fix(ffi_engine): OverlayedDbProvider::succ() must consult overlay, not just snapshot
04-21 16:00	—	xrpl-validator	`6dd8f22`	fix(ffi_engine): tefBAD_LEDGER recovery — fetch missing SLE and retry apply
04-21 16:29	—	xrpl-validator	`8176931`	feat(dashboard): /logs page — progress journal + live GitHub changelog
04-21 16:32	—	xrpl-validator	`a59ae04`	docs(progress): redact file:line for unfixed SEV findings

Committed	Merged into main	Repo	Commit	Subject
04-21 16:42	—	xrpl-validator	`4ab09ec`	fix(ffi_engine): loop tefBAD_LEDGER recovery for multi-missing-SLE owner dirs

Times in UTC. A commit appears on the day it was committed to main or merged into it.